Test 3 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"During summer vacation, Maya joined a community gardening project. She learned to plant vegetables and care for them. Her team also organized workshops to teach younger children about sustainable farming and nutrition. By the end of the program, Maya felt proud of contributing to her community and promoting healthy living."

a) What project did Maya participate in during summer?
b) List two activities Maya performed while volunteering.
c) How did Maya feel after completing the program?
d) Summarize the passage in four sentences using your own words.

2. Grammar & Vocabulary
a) Combine the sentences using an appropriate conjunction:
"It was raining heavily. We decided to stay indoors."

b) Identify the adjectives in the sentence:
"The colorful butterflies flew gracefully over the blooming flowers."

c) Correct the error in this sentence:
"Neither of the boys are willing to help in the kitchen."

d) Convert the following into indirect speech:
"She said, 'I will complete my homework before dinner.'"

3. Writing
Write a short essay (120–150 words) on the topic: *"The Role of Technology in Education"*.
Include advantages, challenges, and personal opinion.

Section B: Mathematics

1. Algebra
a) Solve for x: 5x−12=3x+85x - 12 = 3x + 85x−12=3x+8
b) Factorize: x2−9x+20x^2 - 9x + 20x2−9x+20
c) Simplify: 7p2q−4pq2+3p2q7p^2q - 4pq^2 + 3p^2q7p2q−4pq2+3p2q

2. Geometry & Mensuration
a) The base of a triangle is 15 cm and the height is 12 cm. Find the area.
b) A cylinder has radius 5 cm and height 14 cm. Find the volume.

3. Ratio, Proportion & Percentage
a) A smartphone originally costs $400. It is sold at a 15% discount. Find the selling price.
b) If 8 pens cost $24, find the cost of 20 pens.

4. Data Handling
The marks of 15 students in a mathematics test are: 12, 15, 14, 16, 18, 15, 14, 17, 16, 19, 18, 16, 15, 17, 16.
a) Find the mean, median, and mode.
b) Represent the data using a simple frequency table.

Section C: Science

1. Biology
a) What are the main functions of the human kidneys?
b) Explain photosynthesis in plants with the chemical equation.
c) List two differences between mammals and birds.

2. Physics
a) A car accelerates from 0 to 60 m/s in 12 seconds. Calculate its acceleration.
b) Define energy and provide an example of kinetic energy.
c) Explain the difference between conduction and convection.

3. Chemistry
a) Identify elements, compounds, or mixtures:

- **Sodium chloride**
- **Oxygen**
- **Air**

b) Write a balanced chemical equation for the reaction between copper and nitric acid.

4. Environmental Science
a) List two air pollutants and their effects.
b) Suggest three measures to conserve water at home and in school.

Section D: Social Studies

1. History
a) Name two major revolutions of the 18th century and their outcomes.
b) Explain the significance of the printing press in European history.
c) Identify the countries forming the Allied Powers during World War II.

2. Geography
a) Define latitude and explain how it affects climate.
b) Name the largest desert in Asia and the continent it is located in.
c) Discuss two consequences of deforestation on the environment.

3. Civics / Political Science
a) Distinguish between federal and unitary forms of government.
b) State two responsibilities of citizens in a democracy.

4. Economics
a) Explain supply and demand with examples.
b) Differentiate between fixed and variable costs in business.

Section E: ICT / Computer Studies

1. Computer Basics
a) What is a network? Give one type of computer network.
b) Name one output device and one secondary storage device.

2. Applications
a) Which spreadsheet function finds the highest value in a dataset?
b) Give two uses of presentation software in schools.

3. Programming Basics
a) Define a loop and give an example in Python.
b) Write a Python code snippet to print the multiplication table of 7.

4. Cyber Awareness
a) Explain malware and one method to protect your computer from it.
b) Mention two ways to ensure personal information safety online.

Section F: Islamiyat

1. Quran & Hadith
a) Name two moral values taught in the Quran.
b) Narrate a Hadith that encourages helping others.

2. Aqidah / Belief
a) List the five pillars of Iman (faith).
b) Explain the significance of Tawheed (oneness of Allah) in Islam.

3. Fiqh / Worship
a) What is the importance of Salah in a Muslim's life?
b) Describe the role of Sawm (fasting) during Ramadan.

4. Islamic Ethics & Morality
a) How does Islam encourage honesty in daily transactions?
b) Explain the significance of taking care of the environment according to Islam.

Section G: Arts & Music

1. Visual Arts
a) Explain the concept of perspective in drawing.
b) Name two primary colors.

2. Music
a) Define melody and harmony.
b) Give an example of a string instrument from any culture.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises to improve flexibility.
b) List one professional sport and one recreational activity.

2. Health & Hygiene
a) Why is a balanced diet important for teenagers?
b) Mention two practices to prevent communicable diseases.

3. Safety & Environment
a) State one safety rule during outdoor sports.
b) Suggest one method to reduce household water consumption.